



Making RAG Systems Domain-Specific by Fine-Tuning Embedders on Synthetic Queries

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Introduction

PROBLEM: The quality of any RAG system is bounded by its retriever. If the passage that contains the answer is not retrieved into the top- k , no amount of generation skill can recover the correct response – retrieval failure is an upper bound on end-to-end accuracy.

AIM: We generate a synthetic, in-domain training set from fitness literature, then fine-tune embedding retrievers of different sizes on it and measure the improvement over their off-the-shelf baselines.

Methodology

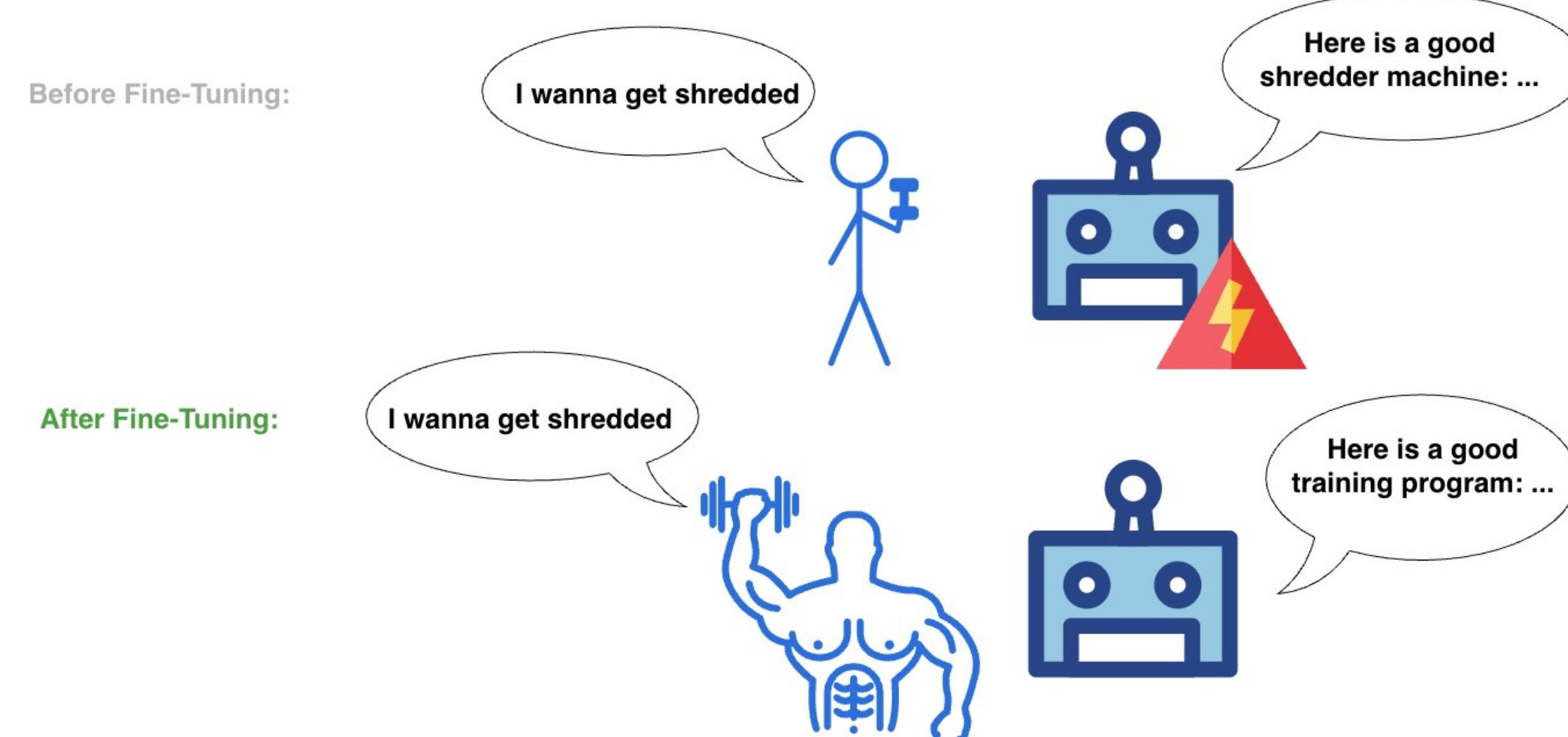
A) Benchmark Creation

- 1 Parse each document into self-contained statements (propositions).
- 2 Group consecutive statements into semantically coherent, variable-length chunks.
- 3 Generate two queries per chunk (formal + informal); a judge verifies each and regenerates failures with feedback.
- 4 Hybrid search (dense + keyword/BM25) retrieves candidate chunks for each query.
- 5 A judge labels each candidate **positive**, **hard-negative**, or **irrelevant**.

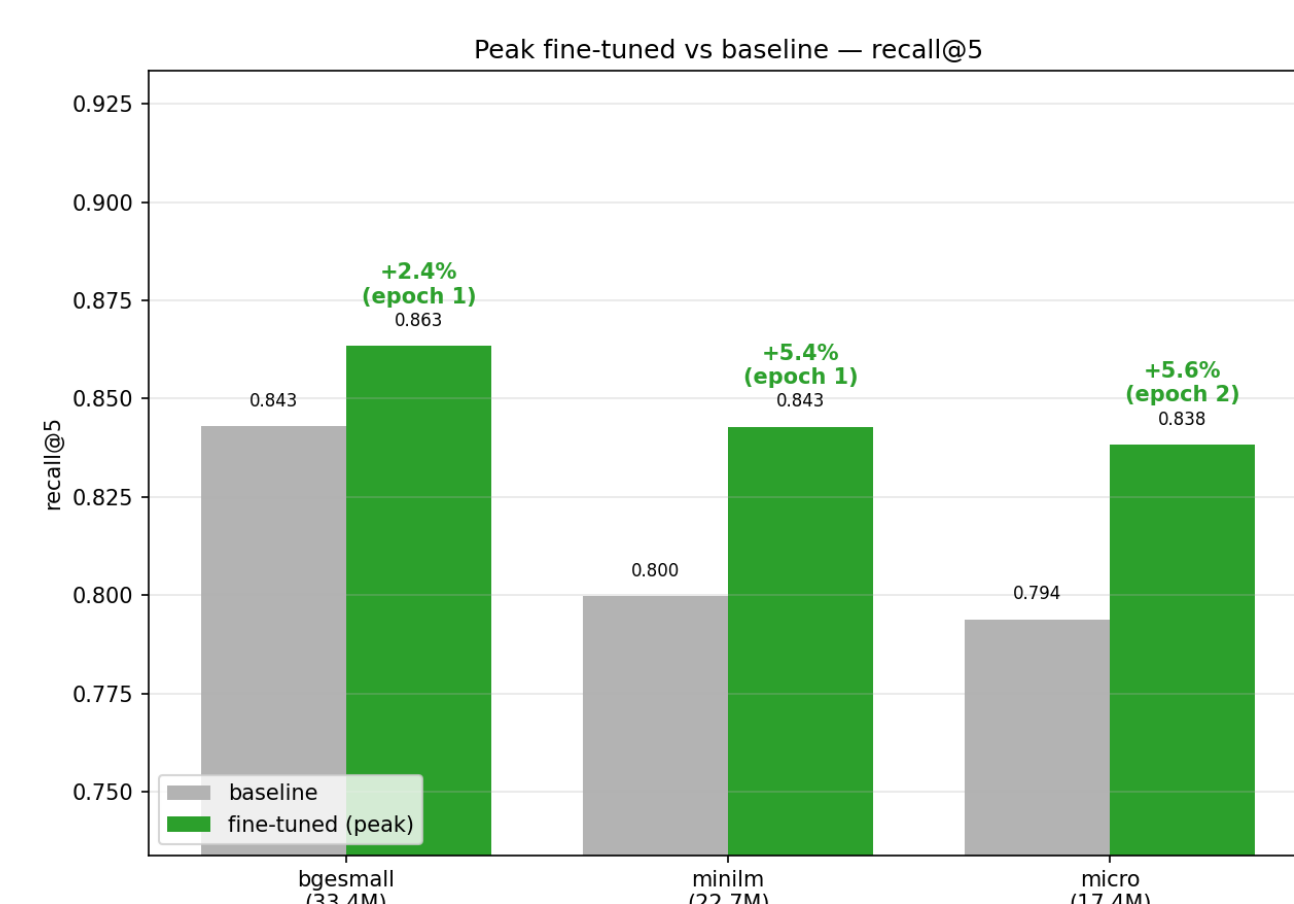
B) Fine-Tuning

- 6 Train the embedder on (**query**, **positive**, **hard-negative**) triplets – positives pull closer, hard-negatives push away. Split by query to prevent leakage.

What We Do

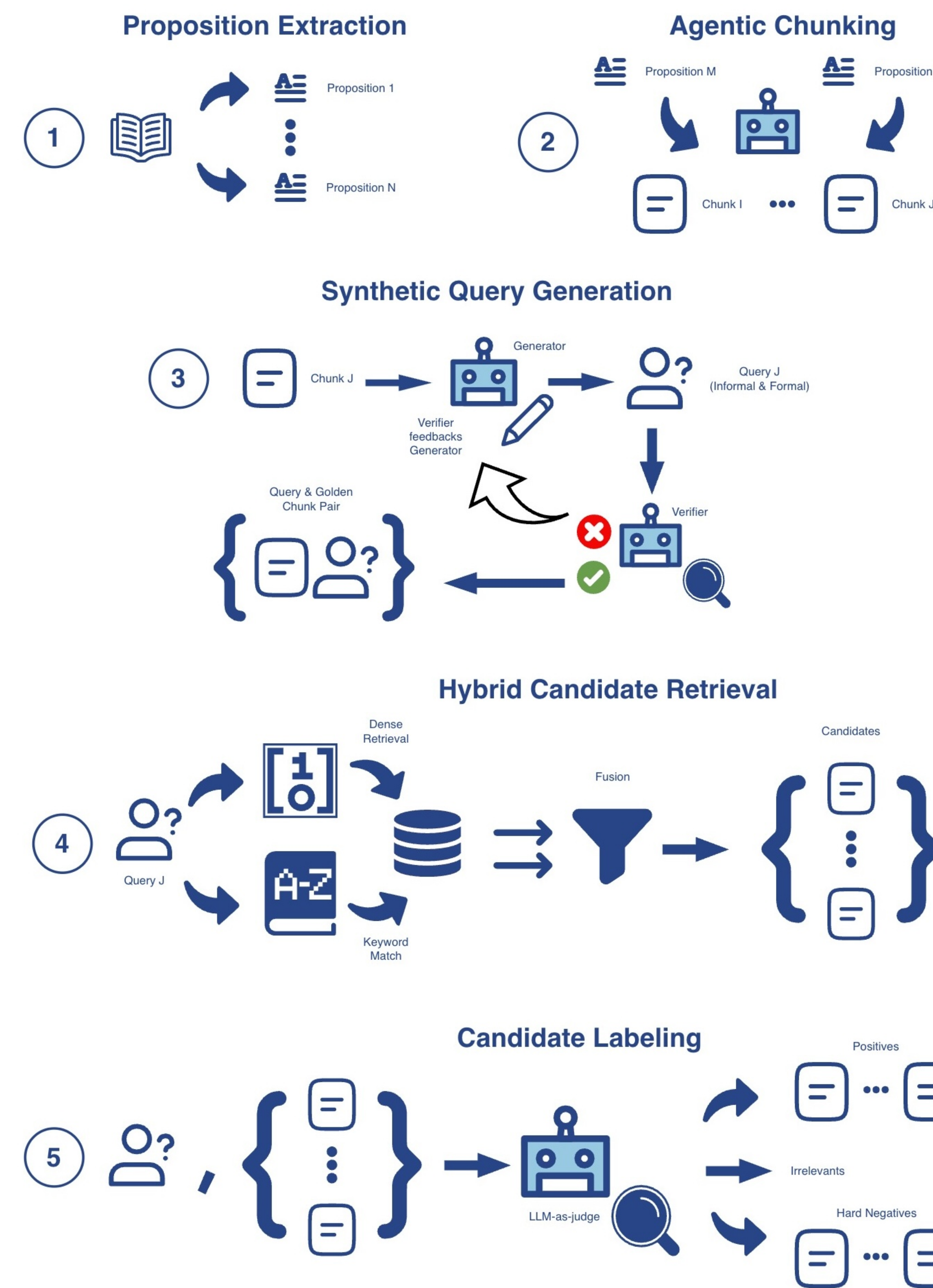


Results



Peak fine-tuned vs. baseline (Recall@5): every model improves.

Pipeline Overview



Models & Setup

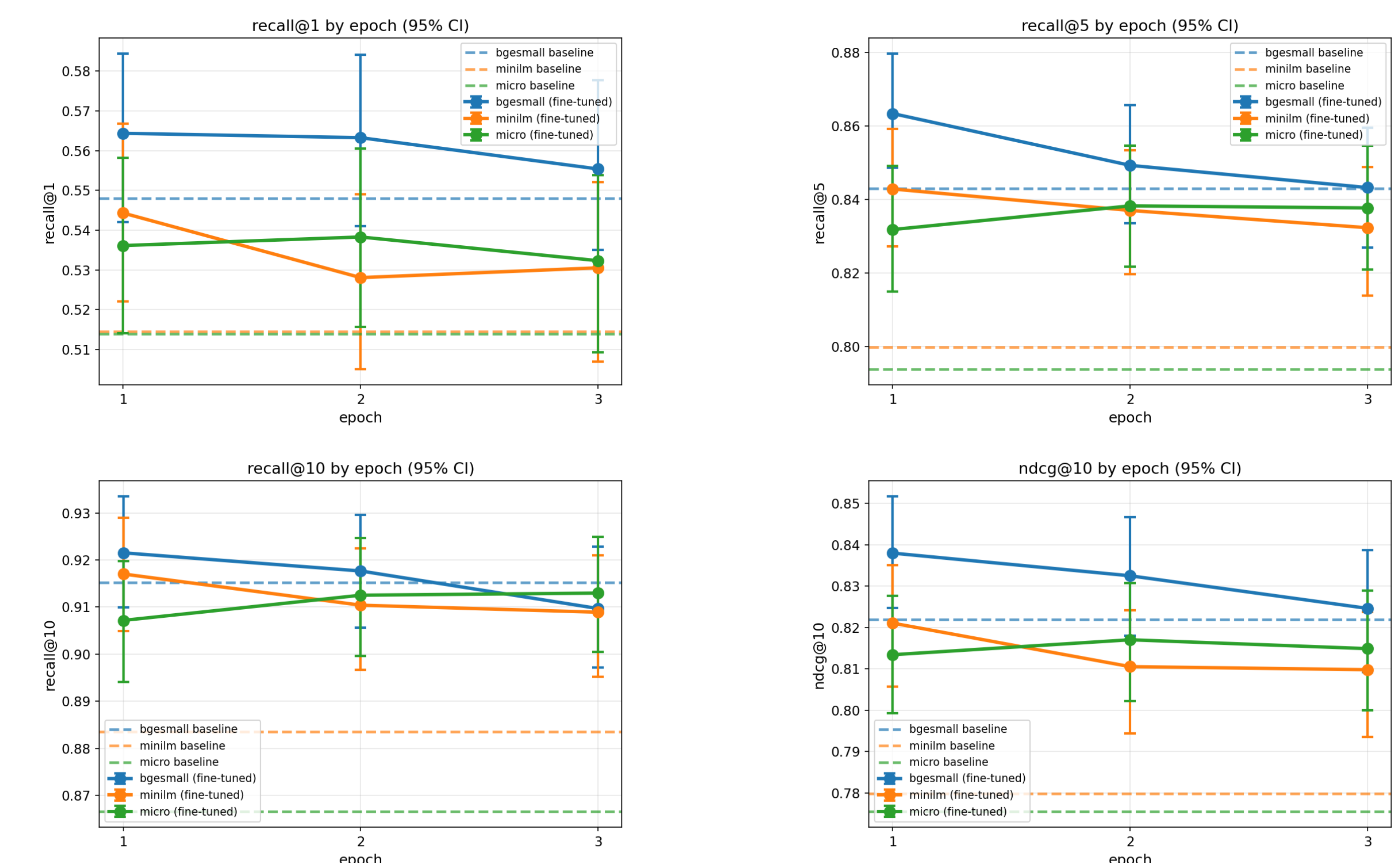
Hardware. Data generation on **NVIDIA L4 (24 GB, GCP)** via vLLM; fine-tuning on a local **Apple M1 (16 GB)**.

Pipeline. Generator **Gemma 2 9B Instruct** (4-bit); judge **DeepSeek-R1-0528-Qwen3-8B** for reasoning-based hard-negative validation.

Retrievers. bge-small (33.4M), all-MiniLM-L6-v2 (22.7M), bge-micro (17.4M).

Training & eval. MNRL loss, batch 16, lr 2e-5, 1–3 epochs; Recall@1/5/10 and NDCG@10 with 95% CIs.

Results (cont.)



Conclusions

- Fine-tuning on synthetic in-domain triplets **beats the off-the-shelf baseline for every retriever**.
- Smaller, weaker bases gain the most (more headroom): bge-micro cuts top-5 retrieval misses ~21% and top-10 misses ~35%.
- Optimal training is short: bge-small and all-MiniLM peak at epoch 1, bge-micro at epoch 2; beyond the peak they overfit.

Future Work

- **Standardized benchmark:** evaluate on a public suite (e.g. BEIR) to measure out-of-domain forgetting.
- **Multi-hop questions:** extend the data and metrics to queries that need several passages.
- **Early stopping & regularization:** train longer without overfitting past the peak epoch.

References

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